

LinkedIn Post Studio — AI-Powered Build Prompt (2026 Edition)

Updated May 2026 to use the latest stable versions of every framework and library.

PROMPT START

Build a production-ready web app called "**LinkedIn Post Studio**" — an AI-powered design tool that helps AI/ML/coding creators generate beautiful, scroll-stopping LinkedIn graphics in seconds.

Project Overview

A single-page web application where users can quickly create professional LinkedIn post images optimized for AI/ML/coding content. Think *"Canva meets vidIQ thumbnail maker"* but specifically for LinkedIn tech creators.

Tech Stack (use exactly this — latest stable versions as of May 2026)

Library	Version	Notes
Next.js	16.2.x (App Router)	Turbopack default, async Request APIs
React	19.x	No forwardRef, use ComponentProps
TypeScript	5.x (strict mode)	No any types anywhere
Tailwind CSS	v4.1.x	CSS-first config via @theme, no tailwind.config.js
shadcn/ui	CLI v4.x, new-york style	Uses unified radix-ui package, OKLCH colors, data-slot attributes
Zustand	v5.x	For global state
Lucide React	latest	For icons
@anthropic-ai/sdk	v0.92.x	Server-side AI calls
Claude model	claude-opus-4-7	Latest Opus 4.7 (May 2026)

 Important Tailwind v4 notes:

- No `tailwind.config.js` — define theme tokens in `globals.css` using `@theme { ... }`
- Single import: `@import "tailwindcss";` (no more `@tailwind base/components/utilities`)
- HSL colors converted to **OKLCH** automatically by shadcn

⚠ Important React 19 / shadcn notes:

- Components no longer use `React.forwardRef` — use `React.ComponentProps<...>` and pass `ref` as a prop
- Every primitive ships with a `data-slot` attribute for styling
- Use `sonner` for toasts (the old `toast` component is deprecated)

Render all post designs as **inline SVG** (NOT canvas, NOT raster images).

📁 Required Folder Structure

```
app/  
  api/  
    generate-thumbnails/route.ts  
    generate-caption/route.ts  
    explain-code/route.ts  
  layout.tsx  
  page.tsx  
  globals.css      # Tailwind v4: @import "tailwindcss"; + @theme {...}  
  components/  
    studio/  
      StudioLayout.tsx  
      ControlPanel.tsx  
      PreviewPanel.tsx  
      DownloadButton.tsx  
    templates/  
      ThumbnailTemplate.tsx  
      ConceptTemplate.tsx  
      FormulaTemplate.tsx  
      CompareTemplate.tsx  
      DayLogTemplate.tsx  
      CurveTemplate.tsx  
      NetworkTemplate.tsx  
      CodeTemplate.tsx
```

```
    QuoteTemplate.tsx
    TipsTemplate.tsx
    AnnounceTemplate.tsx
    ExplainerTemplate.tsx
    CarouselTemplate.tsx
controls/
    TemplatePicker.tsx
    ThemePicker.tsx
    ContentEditor.tsx
    ImageUploader.tsx
    ProfileUploader.tsx
    AuthorInfo.tsx
    QuickAIPanel.tsx
shared/
    PostFooter.tsx
    PostBackground.tsx
ai/
    CaptionGenerator.tsx
ui/                # shadcn components (auto-generated, new-york style)
lib/
    api/claude.ts
    utils/
        svg-export.ts
        text-wrap.ts
        file-upload.ts
    prompts/
        thumbnail.ts
        caption.ts
        code-explainer.ts
    constants/
        themes.ts
        templates.ts
    store/
        useStudioStore.ts
hooks/
    useImageUpload.ts
    useDownload.ts
    useThumbnailGenerator.ts
    useCaptionGenerator.ts
types/
    studio.ts
```

1. Quick AI Thumbnail Mode (primary feature)

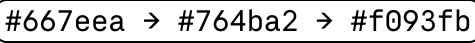
- User types one short idea (e.g., "*Why squared error fails for logistic regression*")
- Click button → AI generates **4 different thumbnail variations** in 4 styles:
 - **Shock** — ALL CAPS, urgent, red warning vibe
 - **Question** — Provocative question, giant (?) decoration
 - **Stat** — Massive number/multiplier (10x, 90%) at center
 - **Reveal** — Personal story format
- Each variant shown as a clickable card; user picks favorite
- Optional fine-tuning fields (hook, subline, context)

2. ML/AI Templates (specialized for tech content)

- **ML Concept** — Big concept name + category badge + definition card + 3 numbered key points
- **Formula** — Beautiful equation display (e.g., $J(\theta) = -[y \cdot \log(h) + (1-y) \cdot \log(1-h)]$) with floating math symbols (Σ , ∂ , f , θ) as decoration
- **VS Comparison** — Side-by-side cards (red ✖ vs green ✔) with bullet points
- **Day Log** — #LearningInPublic format with MASSIVE day number, progress bar, today's focus, key insight quote
- **Curve** — Plotted function curves (sigmoid, ReLU, tanh, gaussian, log loss, convex) with grid background, axes, glow effect
- **Neural Network** — Architecture diagram. User inputs layer sizes (e.g., "4, 6, 6, 3, 1") and it draws connected nodes with color-coded layers (input/hidden/output)

3. Basic Templates

- **Code** — macOS-style window with syntax highlighting (or uploadable screenshot replacement)
- **Quote** — Big serif quote with decorative " "
- **Tips** — Numbered list cards (1-5)
- **Announce** — Big bold launch announcement
- **Explainer** — Code + AI-generated explanation points
- **Carousel** — Multi-slide post (up to 10 slides) with cover, tip, and CTA slides

Theme	Vibe	Colors
Terminal	Hacker	Dark navy + neon green  #00ff88
Cosmic	Bold	Purple-pink gradient  #667eea → #764ba2 → #f093fb
Ocean	Calm	Blue gradient
Sunset	Warm	Red-orange
Minimal	Clean	White + black
Paper	Academic	Cream/yellow

Each theme is a TypeScript object with: `swatch`, `bgGradient[]`, `text`, `muted`, `accent`, `accent2`, `cardBg`, `isDark`.

5. Image Upload Features

- **Profile photo upload** — Replaces default initial circle with circular avatar in footer
- **Hero image upload** — 4 layout modes: Hero (top banner), Split (right half), Background (subtle backdrop), Inline (centered circle). Includes opacity slider.
- **Code block image upload** — Replaces text-based code with uploaded screenshot in macOS-style window frame

6. AI Caption Generator

- Below the preview, generates LinkedIn captions matching the graphic content
- Tone selector: **Professional, Casual, Bold, Storytelling**
- Format: Hook → Short paragraphs with → arrows → Key Takeaways → Engagement question → 5-8 hashtags
- One-click copy to clipboard

7. Author Footer (on every design)

- Bottom-left: Profile photo (or initial circle) + Name + Title/Tagline
- Bottom-right: LinkedIn `in` badge in official blue  #0A66C2
- Horizontal divider line above

- Convert SVG → PNG via Canvas API
 - **1200×1200** dimensions (LinkedIn square)
 - Use **base64 data URLs** (NOT blob URLs) for iframe sandbox compatibility
 - Right-click "Save image as" as fallback
-

🎨 Design Specifications

SVG Canvas

- Every design renders inside a `1200×1200` viewBox SVG
- Background uses `<linearGradient>` with theme's `bgGradient` array
- Subtle grid pattern overlay (60×60 grid, 4% opacity)
- Two radial glow circles in opposite corners (top-left + bottom-right) using accent colors

Typography

- Body: `system-ui, -apple-system, sans-serif`
- Quotes: `Georgia, serif`
- Formulas: `'Cambria Math', Georgia, serif italic`
- Code: `ui-monospace, 'SF Mono', Monaco, monospace`
- Big titles: 60-140px, font-weight 800-900, letter-spacing -1 to -4

Layout Pattern

- 1200×1200 viewBox
 - Standard padding: 80px from edges
 - Footer area: y=1060 to y=1200
 - Content area: y=120 to y=1040
-

🤖 AI Prompts (for Claude API)

Use the official TypeScript SDK pattern:

```
ts
```

```
import Anthropic from '@anthropic-ai/sdk';

const client = new Anthropic({ apiKey: process.env.ANTHROPIC_API_KEY });

const message = await client.messages.create({
  model: 'claude-opus-4-7',
  max_tokens: 1024,
  messages: [{ role: 'user', content: prompt }],
});
```

Thumbnail Generation Prompt:

You are a LinkedIn thumbnail designer for AI/ML/coding content. User idea:
"\${idea}"

Generate 4 scroll-stopping thumbnail variations. Return ONLY valid JSON:

```
{
  "variants": [
    { "style": "shock",    "hook": "...", "subline": "...", "context": "..." },
    { "style": "question", "hook": "...", "subline": "...", "context": "..." },
    { "style": "stat",     "hook": "...", "subline": "...", "context": "..." },
    { "style": "reveal",   "hook": "...", "subline": "...", "context": "..." }
  ]
}
```

Rules:

- "shock": ALL CAPS, 3-5 powerful words, urgent
- "question": Provocative question
- "stat": Big number/multiplier with context
- "reveal": Personal/story-driven
- Hook under 35 chars, subline under 30 chars, context under 25 chars
- Use power words: STOP, NEVER, WHY, HOW, SECRET, TRUTH, MISTAKE
- Be specific to the topic

Caption Generation Prompt:

You are a LinkedIn growth expert for AI/ML creators. Write a high-engagement caption.

Tone: `#{tone}`

Author: `#{handle}`

Graphic content:

`#{content}`

Style:

- Strong hook in first line
- Short paragraphs (1-2 sentences) with line breaks
- Use `→` for bullet points (not `•` or `-`)
- Include "Key Takeaways" section
- End with engagement question
- 5-8 hashtags at end
- 200-400 words

Output ONLY the caption.

Architecture Best Practices

1. **API key on server only** — Never expose `ANTHROPIC_API_KEY` to the client. All AI calls go through Next.js Route Handlers (`app/api/.../route.ts`).
 2. **Single source of truth** — All app state in Zustand v5 store, no prop drilling.
 3. **Pure components** — Templates take props (`theme`, `content`) and return SVG. No internal state.
 4. **Type-safe end to end** — Discriminated unions for templates and styles.
 5. **Custom hooks** — `useImageUpload`, `useThumbnailGenerator`, `useCaptionGenerator` for reusable logic.
 6. **Class-based utilities** — `SvgExporter` class for download logic (testable in isolation).
 7. **One file = one responsibility** — Each template is its own file, registered in a central registry.
 8. **React 19 patterns** — No `forwardRef`. Pass `ref` as a regular prop. Use `React.ComponentProps<'div'>` instead of `HTMLAttributes`.
 9. **Async Request APIs** — In Route Handlers, remember `cookies()` and `headers()` are now async. Always `await` them.
-

Setup Commands

bash

1. Scaffold Next.js 16.2 project (Turbopack on by default in dev)

```
npx create-next-app@latest linkedin-post-studio \  
  --typescript --tailwind --app --turbo --src-dir=false \  
  --import-alias "@/*"
```

`cd` linkedin-post-studio

2. Core dependencies

```
npm install lucide-react zustand @anthropic-ai/sdk sonner
```

3. shadcn CLI v4 – initialize with new-york style + Tailwind v4

```
npx shadcn@latest init
```

4. Add only the components we need

```
npx shadcn@latest add button input textarea label card tabs slider sonner
```

Environment Variables

Create `.env.local`:

bash

```
ANTHROPIC_API_KEY=sk-ant-api03-your-key-here
```

Tailwind v4 — `app/globals.css`

CSS

```
@import "tailwindcss";

@theme {
  --color-brand: #0A66C2;
  --font-display: ui-sans-serif, system-ui, sans-serif;
  --radius-card: 1rem;
}

/* Optional dark mode tokens via @custom-variant */
@custom-variant dark (&:where(.dark, .dark *));
```

No `tailwind.config.js` file is needed. All theme tokens live in CSS via `@theme`. shadcn's `init` will set up the rest (OKLCH base colors, CSS vars, etc.).

🎯 UI Layout

Two-column layout:

- **Left (400px):** Scrollable controls panel with cards for: Templates (grouped: Quick AI / ML/AI / Basic), Color Themes (6 swatches), Content Editor (changes per template), Image Upload, Profile Photo, Author Info
- **Right (flex-1):** Sticky preview with 1200×1200 SVG (rendered in `<svg viewBox="0 0 1200 1200">`), AI caption generator below

Mobile: Stack vertically — controls on top, preview below.

Use rounded corners (`rounded-2xl`), soft shadows (`shadow-sm`), light borders (`border-slate-200`), and a clean white card aesthetic for the controls panel.

🚀 What to Build

Generate the complete codebase following this structure. Start with:

1. Set up the project structure and types
2. Build the Zustand v5 store
3. Implement the `SvgExporter` utility
4. Build 2–3 templates first (`ThumbnailTemplate`, `ConceptTemplate`, `DayLogTemplate`)
5. Wire up the API routes for Claude (`claude-opus-4-7`)
6. Build the UI controls (shadcn new-york style, Tailwind v4)

7. Connect everything

Please write **production-quality, well-commented TypeScript** code. Use proper error handling, loading states, and disabled states. Make the UI polished with smooth transitions and hover effects. Every interaction should feel snappy and intentional.

When you're done, provide:

1. Full setup instructions
2. How to run the dev server (`npm run dev`) — uses Turbopack by default in 16.2)
3. How to deploy to Vercel
4. Screenshot/preview descriptions for each template

PROMPT END

Tips for Using This Prompt

With Claude (claude.ai or API)

- Paste this entire prompt
- Claude will likely build it in stages — you'll need 2-3 follow-up messages like *"continue"* or *"now build the templates"*
- **Best results:** *"Build this step-by-step. Start with the file structure, then types, then store, then templates one at a time."*

With Cursor / Windsurf / Claude Code

- **Best option for full project generation**
- Open empty folder, paste prompt in chat
- Agent will create all files automatically
- Follow up with: *"Run the dev server and fix any errors"*

With v0.dev (Vercel)

- Excellent for the UI parts — paste sections at a time
- Use it for: Studio Layout, Control Panel UI, Template Picker

With Bolt.new / Replit Agent

- Paste full prompt — they'll auto-deploy and run it
 - Good for instant working preview
-

Quick Follow-up Prompts

If the AI gets stuck or produces incomplete code:

- *"Now build the [TemplateName] component with full SVG implementation."*
- *"Add the AI caption generator with proper loading and error states."*
- *"Implement the image upload with size validation and base64 conversion."*
- *"Make the SvgExporter class with PNG download and iframe-safe data URLs."*
- *"Add a profile photo uploader with circular crop preview."*
- *"Wire up the Zustand v5 store with all the state slices."*
- *"Create the Next.js 16 Route Handler for thumbnail generation with proper error handling."*
- *"Convert this component to React 19 patterns — remove forwardRef, use ComponentProps, pass ref as a regular prop."*
- *"Migrate this Tailwind v3 config into Tailwind v4 `@theme` syntax in globals.css."*